Identifying Toxic Language with a Machine Learning Model

IMT 574 Final Project
Authors: Thomas Chang, Rachel Perry

BACKGROUND

This report explores the machine learning models and techniques behind efforts to recognize and classify the toxicity of text excerpts. Here, 'text excerpts' refers to comments submitted to Wikipedia pages and Twitter posts (tweets). This data set was obtained from a kaggle Competition, "Toxic Comment Classification Challenge" that contains both test and training data sets. The training data set contains 6 label categories: toxic, severe toxic, obscene, threat, insult, identity hate. For the purposes of this report we will only focus on the most general "toxic" label but it is important to note that the presence of six unique labels shows that this analysis can go beyond a binary to show the degree and nature of a comment's toxicity as applicable.

The aim of this report is to build a successful model, trained on the Wikipedia text comments, that can then be used to accurately predict the toxicity classification of novel text excerpts. This is an especially important and relevant task given the ubiquity of social media and an ever increasing ability to make statements on the internet with the confidence of anonymity. Wikipedia is one of the most widely used websites that functions as an open-source encyclopedia. Where wikipedia tends to be a great source for historic and static information, Twitter has now become a platform to witness news in real-time. Although these are two separate platforms and may inspire different types or styles of comments, the presence of toxic comments permeates the internet and both platforms. Additionally, because Wikipedia comments are in reaction to something static, the content is potentially more thought out, so toxicity may be more subtle and less explicitly expressed.

Twitter, on the other hand, is built for short form and immediate reactions. Although people can curate longer threads of information on Twitter, the platform's functionality allows for its users to publish much more inflammatory and reactionary content much quicker. This means that Twitter serves as a much more immediate and accurate thermometer when attempting to gauge the temperature of public discourse and the general public reaction to local or global events. For example, in 2020, "toxic messages amount to 21% of the overall conversation touching on the COVID-19 pandemic... 21 out of 100 tweets in our sample are expected to convey a rude, disrespectful, or unreasonable comment. The percentage of toxic tweets increased after 26 March (25%), when many countries were facing the growing adverse effects of the pandemic and passing measures to confine their populations" (Majó-Vázquez, 2020).

In addition to toxicity online regarding the restrictions set in place in response to the pandemic, because of the lock-down mandates, many more people began to rely on the internet for social functions that were no longer available to them. A majority of the industrial world began working from home, socializing from home, working out from home, and all of these new at-home functions necessitated communication over online platforms. Thus the integration of virtual spaces into everyday life and social interactions became necessary and normalized. One study reported

that they “observed a vast increase in digital communication. The data show that 43% of respondents used text messaging more often. This was followed by an increase in voice calls (36%), social media (35%), and video calls (30%). Almost a quarter of people more frequently used email (24%), and just over one fifth played online games more often (22%). Taking all modes together, 46% of respondents had only increased their digital communication, without decreasing any of the methods.” (Nguyen, 2020). The lockdowns significantly accelerated the trends that had already been occurring in terms of a mass social migration into the metaverse. Additionally, there has also appeared to be an increase of violence in the real world (Hassan, 2019). In response to both of these trends, an increase in use of social media and subsequently in toxic discourse and general violence, the Jigsaw team at Google developed an API, Perspective API, that leverages Machine Learning to rate the toxicity of comments and thus the perceived impact the comment might have on a conversation. This is an important distinction made by Perspective, that their definition of toxic includes the consequences of toxicity (i.e. the likelihood of leaving a discussion), meaning there is also some prediction measure for antisocial behaviors. However, prompting a user to exit a conversation due to toxicity, or being exiled from a platform due to toxicity are not necessarily going to deter these users from social media as a whole, this might just incite them to migrate to newer platforms that provide more of a positive feedback loop, or provide further encryption to prevent their toxic behaviors from detection by toxicity monitors. One notable example of this behavior is in regards to planning the riots on the United States Capitol on January 6th when Facebook users migrated to Telegram to avoid detection.

This report aims to build an effective Machine Learning model to contribute to efforts to better understand the form and mechanisms of toxicity online. Building effective Machine Learning solutions to predict toxicity will provide insight into the key variables that not only produce toxicity but allow it to compound on itself and get out of control to the point where individuals feel unsafe or threatened online. Gaining insight into this process can help us control and moderate online platforms to prevent these types of interactions from getting out of hand and negatively affecting people in the real world.

METHODOLOGY

When considering Machine Learning models for processing text and language, a majority of the detailed work is done when preprocessing the data. The first step is to understand the process of text classification, by which you divide the entire segment of text into its sub parts and categorize the sub parts to gain more insight about the body of the text as a whole. This is done using defined methods outlined by Natural Language Processing, a branch of artificial intelligence, and then using classification algorithms in this case Naive Bayes, Support Vector Machine classifiers to build the classification model. The accuracy and successful classification of these models will then be compared to the result of using a Random Forest classification.

The data used in this report was read into python and distributed into X and y data frames. A random seed was set to ensure reproducibility across test cases. The data was preprocessed by removing any blank instances, changing all the text to lowercase (i.e. so that “WoRd” and “word” are processed as the same unit). Each sample of text was then tokenized, which is the process of taking a body of text and splicing it into its most basic units of meaning (i.e. individual words or

tokens). Each token is then reduced further through a process of lemmatization, in which the original word becomes its common base or root word. In contrast to stemming, lemmatization intends to preserve the context or the meaning that the word provided to the original body of text. This is an important distinction for this report because this linguistic analysis is focused on the valence of the text and the meaning implied through that valence. Although we can categorize the 'toxicity' of comments, this is still a subjective category that can be highly dependent on the context, not only for determining toxic versus non-toxic but also for differentiating between types of toxicity. See figure for an example of the before and after result of pre-processing.

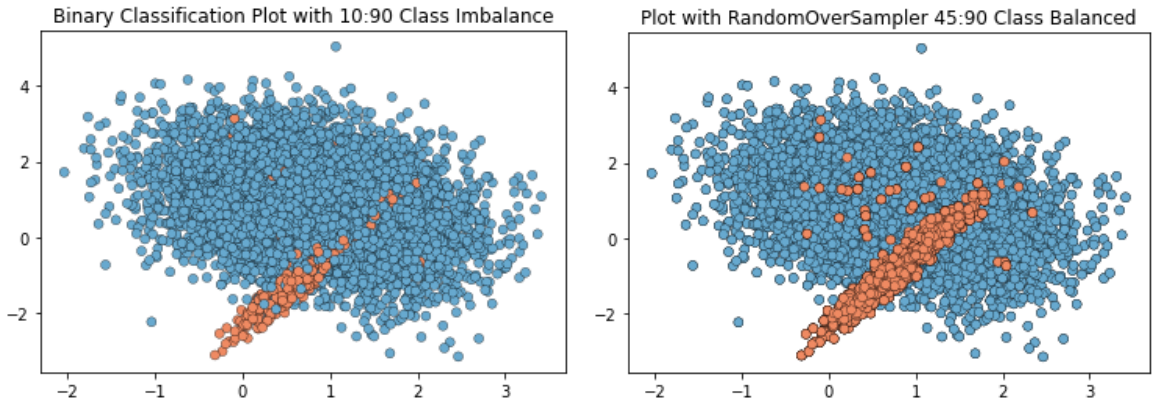
Raw Text	Pre-processed Text
Stuning even for the non-gamer: This sound track was beautiful! It paints the senery in your mind so well I would recomend it even to people who hate video game music! I have played the game Chrono Cross but out of all of the games I have ever played it has the best music! It backs away from crude keyboarding and takes a fresher step with grate guitars and soulful orchestras. It would impress anyone who cares to listen! ^_^	['stun', 'even', 'sound', 'track', 'beautiful', 'paint', 'senery', 'mind', 'well', 'would', 'recomend', 'even', 'people', 'hate', 'video', 'game', 'music', 'play', 'game', 'chrono', 'cross', 'game', 'ever', 'play', 'best', 'music', 'back', 'away', 'crude', 'keyboarding', 'take', 'fresh', 'step', 'grate', 'guitar', 'soulful', 'orchestra', 'would', 'impress', 'anyone', 'care', 'listen']

Figure 1. Raw and Preprocessed Text (Bedi, 2020)

Each of these individual word units are then vectorized, using "Term Frequency - Inverse Document Frequency". This gives each term a frequency score for how often they appear in the dataset as a whole, and accounting for words that appear, as a metric for weight or importance of those word units (i.e. importance within an instance and scaling down words that appear across all instances). This process then builds a vocabulary based on the TF-IDF analysis to assign a value to each of the word units which is then applied to transform the training data based on the vocabulary (see Figure 3 for the word cloud representation of term frequency in the training dataset).

In parallel with the data preprocessing, exploratory data analysis is conducted to understand how to set up the data to perform best on the model. Because the data sets for this report were already split into train and test sets, a train/test split is not a necessary step to prepare the model. The original training dataset from Wikipedia included over 159,000 rows of observations. A sample of 2,000 observations of the training dataset has been taken in order to reduce model runtime from 4 hours down to 1 minute. In addition to this, the training data set has already been encoded with a binary value (i.e. presence or absence, 0 or 1) for each of the six toxicity categories: toxic, severe toxic, obscene, threat, insult, identity hate. For the initial analysis the six metrics were compressed to produce one probability/intensity value for each instance. This processing produced a preliminary model using Naives Bayes with 82% accuracy as compared to an analysis in which only the "toxic" column was used as the training predictor, which predicted with 96% accuracy. Therefore the training data for this report used only the most general "toxic" category for the models.

The last preprocessing technique is to see if there is a class imbalance issue that needs to be solved so that the model is trained on a set with equal representation of toxic and non toxic comments. The training dataset had an imbalance of 9.59% toxic labels and 90.41% non-toxic labels while the test dataset had a class imbalance of 10.67% to 89.33% for toxic to non-toxic labels, respectively.



Counter({0: 8952, 1: 1048})	Counter({0: 8952, 1: 4476})

Figure 2. Class Balance

Over-sampling was performed to raise the number of the minority labels and balance the data. A 50% sampling strategy, as opposed to a 'minority' strategy, was implemented so that there would not be too much overfitting (i.e instead of a 1:1 ratio this provides a 1:2 ratio of toxic to non toxic inputs). Both the imbalanced and oversampled sets will be used in the analysis process to validate the accuracy results. The preprocessing of the data is complete and now the models can be trained using the three separate techniques.

Because this dataset is using labeled data to train the model and is being used to make a prediction on novel data, supervised machine learning techniques will be used. The first model is the Naive Bayes Classifier. The Naive Bayes is particularly successful with NLP problems because it assumes each feature's contribution to the classifier is independent of one another, and because the parameters are estimated using a version of maximum likelihood, or relative frequency counting (leveraging the TF-IDF preprocessing). The underlying algorithm is based on Bayes' Theorem which uses conditional probability to calculate the overall probability of an event. Thus the determination of text being toxic or non-toxic is determined by the probability of each of the token units appearing in a toxic or non-toxic text sample, and the compounded probability of the entire text being toxic given the rest of the words and their frequency calculations.

The second model is a Support Vector Machine. This model is also good with NLP because it calculates hyperplanes within the word vector space to separate the data into distinct categories. The NB and SVM approaches both produce models that are trained based upon how (context and frequency) and where (in which class label) the word units appear to make the prediction for our toxic / non-toxic classifier.

For a point of comparison this report will also use a Random Forest Classification Model. The Random Forest is an ensemble method in which a number of decision trees are generated to make the class distinctions and then the results are averaged across all test instances to make the classification. Random Forest is a reliable modeling technique and will provide a good point of reference to show whether the other techniques that are typically used for NLP problems perform notably better or with the same ability as the Random Forest. All three models will use the default hyperparameters. SVM in particular did not benefit from changing the parameters due to being a binary classification.

The accuracy and efficacy of these models will then be tested using a novel data set acquired directly from Twitter using their API. The most frequent words in the training data set were: article, page, wikipedia, use, make, and one.

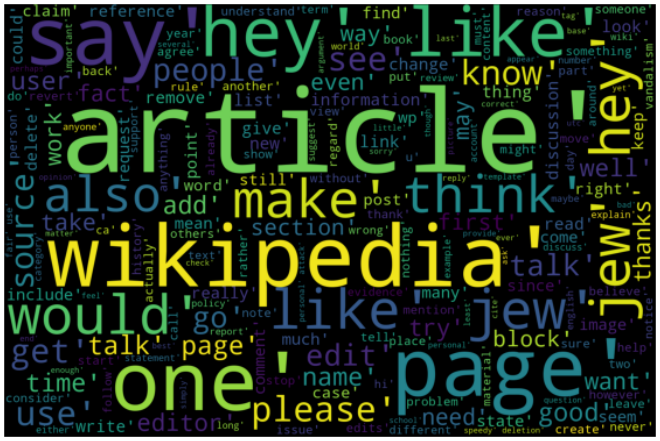


Figure 3. Wikipedia Training Data, Frequency Word Cloud

The query to build the Twitter test set requires a key word, and to make the data sets more comparable than a random query, the words “use”, “make” and “one” were used as opposed to the most frequent words which were very “Wikipedia-specific.” The test set consists of 300 tweets that were scored by the investigators for toxicity. To better define and understand the nuances of toxicity and to increase the inter-annotator agreement for the toxicity classification, definitions for each of the original categories are provided. For the annotations, only the most general category of toxicity was used, but if any of the other category requirements were met that was another automatic indication of toxicity in the scoring process.

Toxic	Severe Toxic	Obscene	Threat	Insult	Identity hate
Negative valence comment, intended to provoke a response.	Extreme negative valence comment, intended to provoke a strong retaliatory response.	Contains a profanity word.	Contains violent language.	Comment that is an attack.	Expresses prejudice against a particular group, especially on the basis of race, religion, or sexual orientation

Figure 4. Definition of Toxicity Classes

ANALYSIS

Once the test set was scored by both annotators, an inter-annotator agreement analysis was conducted to determine the overlap and accuracy of the scores. In the initial analysis, misclassification was a concern when labeling due to inherent bias in what we viewed as toxic. Subtle contextual observations were also considered upon understanding what type of accounts were posting their tweets. In example, sexual innuendos that may not initially appear toxic, especially to the model, were scored differently by the two annotators. Using Cohen’s kappa score, the inter annotator agreement is 73.54%. This is a good level of agreement between annotators that validates the use of both of the labeled datasets for the testing process.

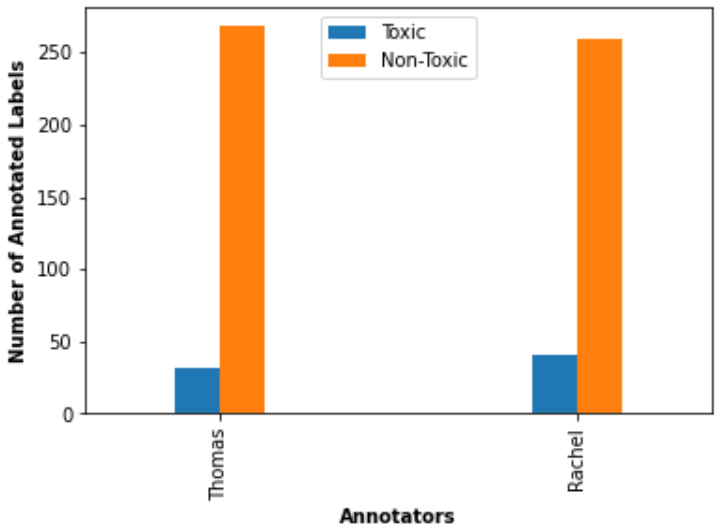


Figure 5. Inter-Annotator Agreement of Class Labels

In the exploratory data analysis stage of the process, the balance of the toxic and nontoxic classes was noted as a 10:90 split, with about 10% of toxic comments in the original set. During the analysis, creating a balanced 50/50 classification resulted in model accuracies below 60%. With a 50% sampling strategy, to bring the toxic to nontoxic classes to a 45:90 split, the results of each of the models tested can be seen below. Even after adjusting the sampling strategy, the balanced (i.e. oversampled) training data still caused the models to perform with less accuracy as compared to the original unbalanced data set. Due to this decrease in accuracy, the final results reported here were trained on the original data set and thus used the unbalanced set.

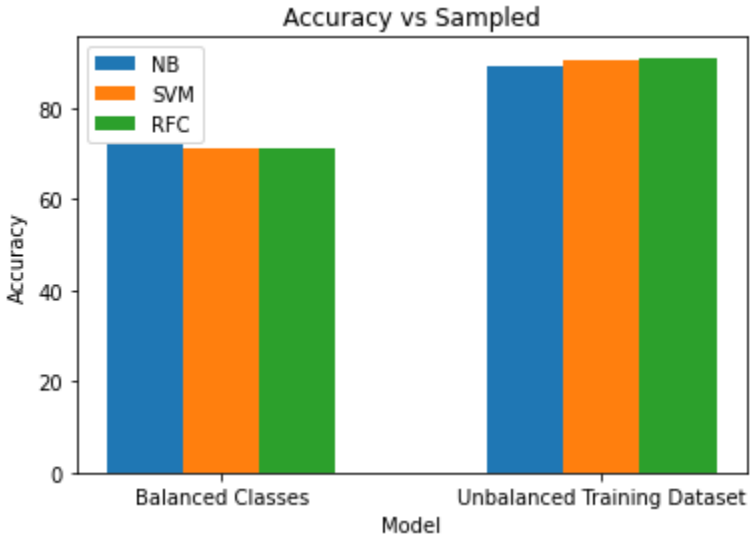


Figure 6. Model Performance Based on Class Balance of Training Data

RESULTS

The results of running the various models on the two annotated test datasets are seen below. Each model was run 5 times per annotator. The resulting accuracy ranges and averages, comparing the balanced and unbalanced classification, are presented below.

Balanced Data				
Type of Model	**Annotator 1 Dataset**		**Annotator 2 Dataset**	
	Range	Average	Range	Average
Naive Bayes	69.90% - 72.42%	71.4%	68.56% - 69.59%	69.07%
Support Vector Machine	68.90% - 74.38%	71.14%	69.32% - 74.23%	70.98%
Random Forest	69.9% - 72.39%	70.81%	64.69% - 72.68%	69.59%
Unbalanced Data				
Type of Model	**Annotator 1 Dataset**		**Annotator 2 Dataset**	
	Range	Average	Range	Average
Naive Bayes	89.33%	89.33%	86.33%	86.33%
Support Vector Machine	89.33% - 90.0%	89.67%	86.33% - 88%	87.33%
Random Forest	90.33% - 90.66%	90.5%	86.33% - 88%	87.16%

Figure 6. Model Performance Based on Class Balance of Training Data, per Annotator

In addition to the accuracy scores, confusion matrices are an important metric here to understand the classification choices of the model, namely if the errors we're producing are false positives or false negatives.

Balanced Data		
Type of Model	**Confusion Matrix**	**Accuracy**
Naive Bayes	[257 11] [106 28]	70.895%
Support Vector Machine	[259 9] [112 22]	69.900%
Random Forest	[265 3] [100 34]	74.378%
Unbalanced Data		
Type of Model	**Confusion Matrix**	**Accuracy**
Naive Bayes	[268 0] [32 0]	89.33%
Support Vector Machine	[267 1] [27 5]	90.67
Random Forest	[265 3] [25 7]	90.67

Figure 6. Model Performance Based on Class Balance of Training Data, Confusion Matrices

CONCLUSION

All of the models performed with a high degree of accuracy, with the decrease due to the balanced class dataset. All three models performed within the same very small margin of accuracy, in the 86-90% range. This high accuracy among all three of the models is very promising because

it shows that the task of identifying toxicity is well suited for machine learning in general and can be performed well with a number of different models, each of which could probably be refined specifically to more niche natural language processing tasks in this domain.

Additionally, all of the models' classification reports show a bias towards false positives. This means that the model is erring on the side of caution and over-assessing toxicity. This has positive implications for the practical application of the model because with such a sensitive topic it would be best practices to err on the side of caution, meaning that if the goal is to monitor and flag toxic content then the model should be trained in a way where errors produce much more false positives than false negatives. In this case a false negative would mean a toxic comment that was not recognized as such and allowed to persist on the online platform, whereas a false positive would be a harmless comment that was labeled as toxic. In these situations, saving a separate censorship conversation, it's safer to flag harmless content as toxic and remove it from general circulation than it is to miss toxic comments.

Overall there is promise in this field in terms of building reliable and strong prediction models for toxicity. Future models could pull data from a number of online resources to get a variety of training data (i.e. a mix of long and short form inputs as well as a variety of formal and colloquial writing). This would allow future models to increase their scope depending on the use case of the project. For example there is a strong future for the Perspective API in building and refining this level of generalizability to be a resource to be widely applicable to any online medium that needs support in monitoring their comments. However there are other use cases where the type of commentary that is being monitored is more niche or the toxicity is purposely veiled and maybe takes on other implicit forms that would require a more extensive feature space with intentional hyperparameters that will contribute to the accuracy of the model.

These improvements to future machine learning models for identifying toxicity show promise for promoting positive online discourse and reducing the human toll that this type of manual monitoring takes. For example, Facebook employees tasked with manually identifying toxic or harmful material that did not adhere to the community guidelines showed increased emotional distress and felt more isolated from friends and family, merely by being exposed to the harmful materials (Newton, 2019). There is a wealth of psychological research on the effects of exposure to violence and violent imagery but we do not yet know if these effects are transferred online when people are exposed to toxic language. However, as good stewards of the online space it should be our responsibility to invest in pro-social tools to help monitor and maintain positive interactions online, which is a body of important work that future toxicity models can contribute to and that this report aims to bring attention to.

RESOURCES

Bedi, G. (2020, July 13). *Simple guide to text classification(nlp) using SVM and Naive Bayes with python*. Medium. Retrieved March 4, 2022, from https://medium.com/@bedigunjit/simple-guide-to-text-classification-nlp-using-svm-and-naive-bayes-with-python-421db3a72d34

Brownlee, J. (2020, February 14). *One-Class Classification Algorithms for Imbalanced Datasets.* MachineLearningMastery. Retrieved March 5, 2022, from https://machinelearningmastery.com/one-class-classification-algorithms/

Dessì, D., Recupero, D. R., & Sack, H. (2021, March 25). *An assessment of deep learning models and word embeddings for toxicity detection within online textual comments*. MDPI. Retrieved March 5, 2022, from https://www.mdpi.com/2079-9292/10/7/779/html

Hanu, L. (2021, February 8). *How AI is learning to identify toxic online content.* Scientific American. Retrieved February 28, 2022, from https://www.scientificamerican.com/article/can-ai-identify-toxic-online-content/

Hassan, A. (2019, November 12). *Hate-crime violence hits 16-year high, F.B.I. Reports.* The New York Times. Retrieved March 4, 2022, from https://www.nytimes.com/2019/11/12/us/hate-crimes-fbi-report.html

Koratana, A., & Hu, K. (n.d.). *Toxic speech detection*. web.stanford.edu. Retrieved March 5, 2022, from https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1194/reports/custom/15744362.pdf

Kshirsagar, V. (2019, December 24). *Detecting hate tweets - twitter sentiment analysis.* Medium. Retrieved March 1, 2022, from https://towardsdatascience.com/detecting-hate-tweets-twitter-sentiment-analysis-780d8a82d4f6

Majó-Vázquez, S. (2020, July 9). *Volume and patterns of toxicity in social media conversations during the covid-19 pandemic.* Reuters Institute for the Study of Journalism. Retrieved February 28, 2022, from https://reutersinstitute.politics.ox.ac.uk/volume-and-patterns-toxicity-social-media-conversations-during-covid-19-pandemic

Newton, C. (2019, February 25). *The trauma floor*. The Verge. Retrieved March 9, 2022, from https://www.theverge.com/2019/2/25/18229714/cognizant-facebook-content-moderator-interviews-trauma-working-conditions-arizona

Nguyen, M. H. (2020, September 9). *Changes in digital communication ... - journals.sagepub.com*. Sage Journals. Retrieved March 6, 2022, from https://journals.sagepub.com/doi/10.1177/2056305120948255?icid=int.sj-related-articles.similar-articles.32

Noever, D. (n.d.). *Machine Learning Suites for Online Toxicity Detection.* https://arxiv.org/. Retrieved March 5, 2022, from https://arxiv.org/pdf/1810.01869.pdf

Using machine learning to reduce toxicity online. Perspective. (n.d.). Retrieved March 5, 2022, from https://www.perspectiveapi.com/how-it-works/

Toxic comment classification challenge. Kaggle. (n.d.). Retrieved March 10, 2022, from https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge/data